NexusMart | Brief Project Documentation

CodeConnect 2026 • Data Analytics Module • Working status as of 18 October 2026

1. Project objective and architecture

Build a reproducible pipeline that transforms nine raw marketplace CSVs into cleaned datasets and a PostgreSQL star schema, orchestrated by Apache Airflow, with an interactive Power BI dashboard and evidence-backed business recommendations.

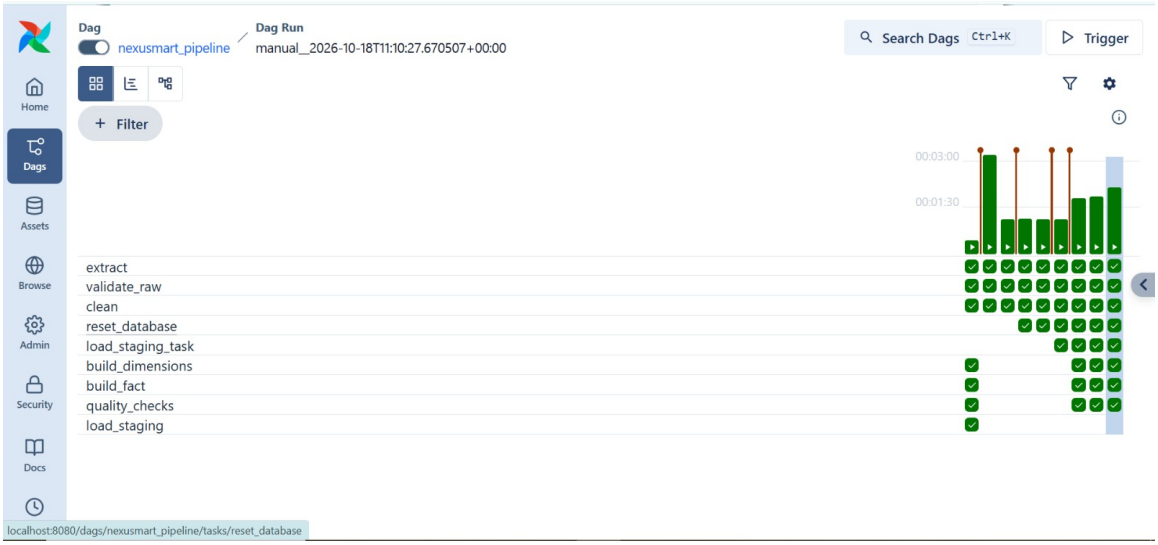
Flow: Raw CSVs → validation → Python cleaning → warehouse reset → PostgreSQL staging → five dimensions → FactSales → automated quality checks → Power BI.

2. Implemented pipeline

Task	Role
extract	Checks presence of nine raw CSVs
validate_raw	Checks CSV headers and nonempty files
clean	Runs include/clean/run_all and writes processed files, rejects and maps
reset_database	Clears staging, fact and non-Unknown dimensions before reload
load_staging_task	Loads the nine cleaned CSVs to PostgreSQL staging
build_dimensions	Populates the five dimension tables
build_fact	Builds FactSales at order-item grain
quality_checks	Checks fact counts, key integrity and fee allocation

The team reported successful Airflow runs. Final reproducibility still requires a second full run with stable counts and saved execution evidence; quality-check coverage must be confirmed against the latest SQL rules.





3. Cleaning results from clean_stats.csv

Dataset	Rows in	Rows after file-level clean	Rejected
Locations	650	650	0
Sellers	420	420	0
Products	3,200	3,097	103
Customers	45,000	42,983	2,017
Orders	163,200	160,000	3,200
Order items	315,143	313,871	1,272
Payments	168,111	164,815	3,296

Shipping	151,228	151,228	0
Returns	16,048	16,048	0

The file-level removals included 2,017 merged customer records, 103 merged product records, 3,200 exact duplicate orders, 3,296 exact duplicate payments, and 1,272 invalid-quantity order lines. Duplicate ID mappings are preserved in customer_map.csv and product_map.csv; rejected records have reason codes.

Additional cross-file cleaning reduced orders to 159,988 (12 unrepaired order dates) and order_items to 313,836 (29 orphan-order lines and 6 invalid-price lines). It also identified 321 orders mapped to Unknown customers, 929 order items mapped to Unknown products, and 210 orphan return rows rejected. These stages explain why final counts differ from file-level rows_out.

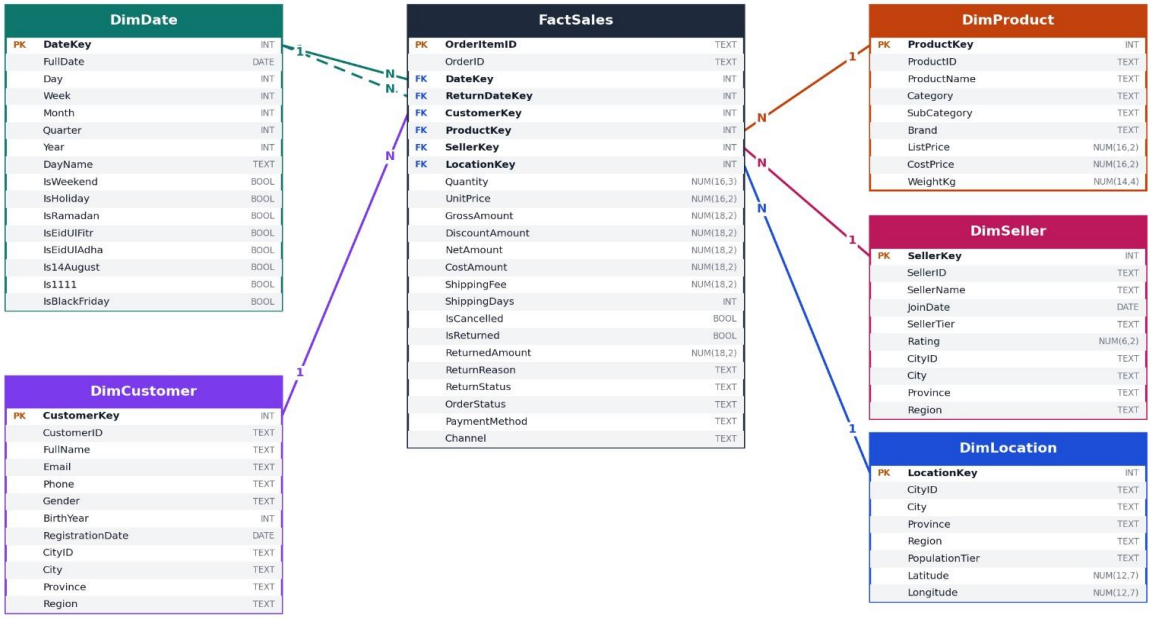
4. Warehouse / star schema

Exactly one fact table (FactSales: one row per order item) and five dimensions (DimDate, DimCustomer, DimProduct, DimSeller, DimLocation). Dimensions link to FactSales, not to each other; missing dimension references use an Unknown member. Shipping fee must be allocated once per order across its line items; valid returns exclude Rejected statuses.

Core SQL assets: 00_reset.sql, 01_schema.sql, load_staging.sql, 02_dimensions.sql, 03_fact_sales.sql, verify_star_v2.sql. The schema diagram should be exported as docs/star_schema.png and verified against final table definitions.

NexusMart Data Warehouse: Star Schema

Grain of FactSales: one row per order line item



Notes

- Dimensions are not linked to each other (single star).
- Dashed line: ReturnDateKey is a second (role-playing) reference to DimDate.
- Each dimension also holds an 'Unknown' member for records without a valid reference.
- Returns are aggregated per order line (valid, non-Rejected returns only).

Legend

PK Primary key (surrogate key)
FK Foreign key to a dimension
1 / N One dimension row to many fact rows

```
 table_name  | row_count
-------------+-----------
 FactSales   |    315109
 DimDate     |       732
 DimCustomer |     42984
 DimProduct  |      3097
 DimSeller   |       421
 DimLocation |       651
(6 rows)
```

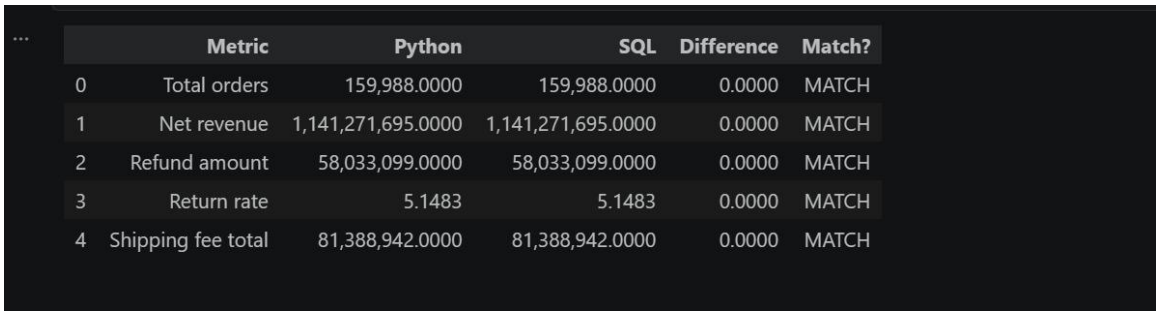
```
nexusmart=# \dt
             List of relations
 Schema |       Name        | Type  | Owner
--------+-------------------+-------+-------
 public | DimCustomer       | table | nexus
 public | DimDate           | table | nexus
 public | DimLocation       | table | nexus
 public | DimProduct        | table | nexus
 public | DimSeller         | table | nexus
 public | FactSales         | table | nexus
 public | customer_map      | table | nexus
 public | pipeline_run_log  | table | nexus
 public | product_map       | table | nexus
 public | rejects           | table | nexus
 public | stg_customers     | table | nexus
 public | stg_locations     | table | nexus
 public | stg_order_items   | table | nexus
 public | stg_orders        | table | nexus
 public | stg_payments      | table | nexus
 public | stg_products      | table | nexus
 public | stg_returns       | table | nexus
 public | stg_sellers       | table | nexus
 public | stg_shipping      | table | nexus
(19 rows)
```

5. Metric reconciliation — current SQL benchmark

Metric	PostgreSQL result
Distinct orders	159,988
Net revenue	PKR 11,412,716,195
Valid refund amount	PKR 58,033,099
Return rate	5.1483%
Allocated shipping fee total	PKR 81,388,942

These figures were reported from the team's PostgreSQL query results. The distinct FactSales order count matched stg_orders (159,988). Pandas-versus-SQL comparison was performed in the notebook, but the final three-layer reconciliation CSV and Power BI cards still need a final validated, unfiltered side-by-side comparison.

Metric	Definition
Total orders	Distinct `order_id` after cleaning, including cancelled orders
Net revenue	Non-cancelled net merchandise value minus valid refunds
Refund amount	Sum of valid refunds, excluding Rejected returns
Return rate	Lines with a valid return ÷ delivered lines
Shipping fee total	Shipping fees counted once per order, not once per line



	Metric	Python	SQL	Difference	Match?
0	Total orders	159,988.0000	159,988.0000	0.0000	MATCH
1	Net revenue	1,141,271,695.0000	1,141,271,695.0000	0.0000	MATCH
2	Refund amount	58,033,099.0000	58,033,099.0000	0.0000	MATCH
3	Return rate	5.1483	5.1483	0.0000	MATCH
4	Shipping fee total	81,388,942.0000	81,388,942.0000	0.0000	MATCH

6. Open data-quality findings and limitations

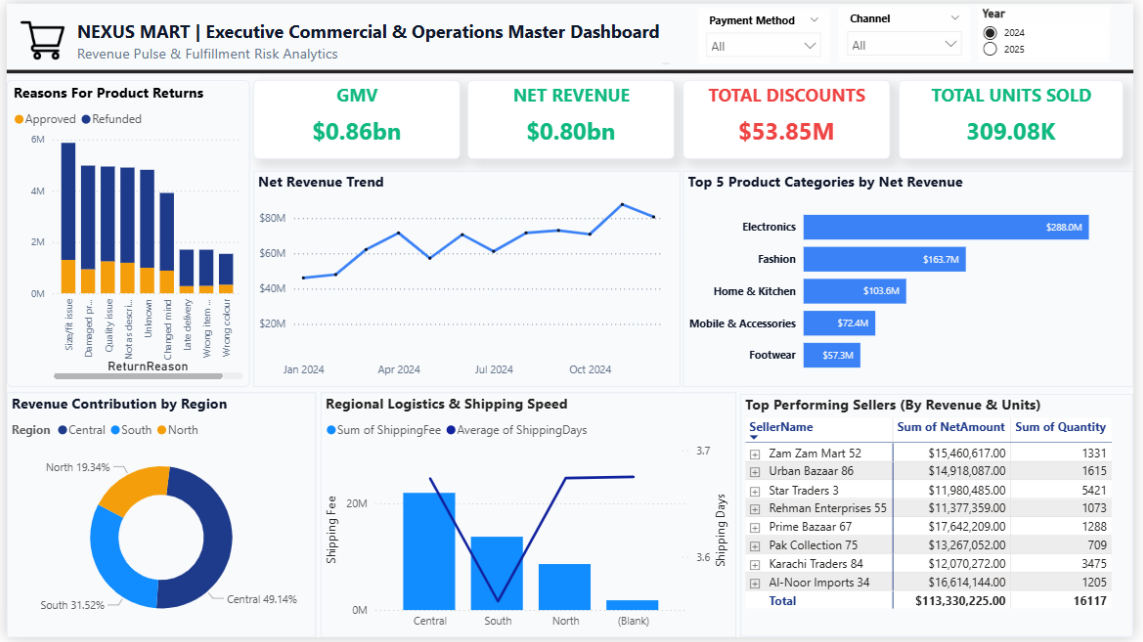
clean_stats.csv records 3,641 orders with totals differing from summed order lines, and 265 orders without any surviving line. These checks are flags, not necessarily confirmed material financial errors; investigate thresholds and business impact. The dataset also contains Unknown customer/product references and imputed discounts or shipping fees. Record source, row population, repair rule and reporting effect for each finding rather than hiding it.

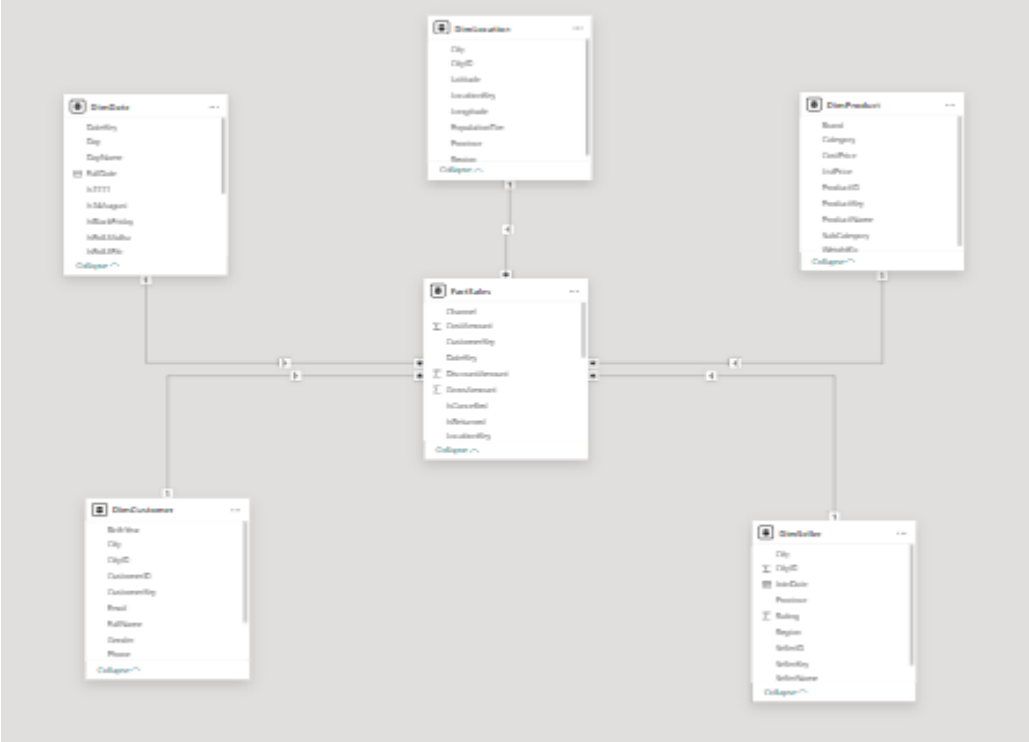
The 17-check verify_star_v2.sql distinguishes FAIL from REVIEW, but it only returns status rows; Airflow must explicitly raise an error on FAIL. Review counts require commentary, not automatic rejection. Avoid claiming all checks passed until executed results are preserved.

7. Power BI and pending deliverables

The dashboard must read PostgreSQL FactSales and the five dimensions (not raw CSVs). It must cover overview, time, products, geography, customers, sellers, and returns/cancellations, with required slicers, drill functionality and a tooltip or bookmark. The Power BI .pbix and PDF export are not yet confirmed complete.

Remaining submission work: (1) complete and refresh dashboard, (2) verify DAX card values against pandas/SQL, (3) finalize 2–3 page executive insights with evidence, (4) prepare README and Data Quality Report, (5) collect DAG and database evidence screenshots, (6) test a fresh reproducible setup and two sequential DAG runs, and (7) package SQL, code, diagram, PBIX, PDF and documentation without secrets.





8. Evidence and source artifacts

Recommended evidence: successful Airflow graph and logs; six PostgreSQL table row counts; schema diagram; clean_stats.csv and rejects.csv; reconciliation table with Python, SQL and Power BI results. Primary project references: CodeConnect Data Analytics Module Problem Statement, NexusMart Datathon Team Workflow, and the team’s processed CSV / SQL artifacts.

Module	File	Notes
locations.py	locations.csv	coordinates outliers, province spellings
sellers.py	sellers.csv	rating 0-5, join_date not limited to order window
products.py	products.csv	price strings ("Rs. 1,250"), duplicate merge, outliers
customers.py	customers.csv	duplicate merge, phone normalisation, birth_year range
orders.py	orders.csv	bad dates kept as NaT for repair, discount_pct from code
order_items.py	order_items.csv	quantity, price, discount
payments.py	payments.csv	duplicate payments, payment_rank
shipping.py	shipping.csv	one shipment per order, fee flag
returns.py	returns.csv	is_valid_return
integrity.py	cross-file	orphans, date repair, discount reconciliation